Guidelines for a successful thesis writing

Dissertation (EI or ETI)

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Guidelines

- **A Rule Of Thumb:**

  Good writing is essential in a dissertation. However, good writing cannot compensate for a paucity of ideas or concepts. Quite the contrary, a clear presentation always exposes weaknesses.

- **Definitions And Terminology:**

  1. Each technical term used in a dissertation must be defined either by a reference to a previously published definition (for standard terms with their usual meaning) or by a precise, unambiguous definition that appears before the term is used (for a new term or a standard term used in an unusual way).

  2. Each term should be used in one and only one way throughout the dissertation.

  3. The easiest way to avoid a long series of definitions is to include a statement: ``the terminology used throughout this document follows that given in [CITATION].'' Then, only define exceptions.

Guidelines

- Terms And Phrases To Avoid:
  - Adverbs: Mostly, they are very often overly used. Use strong words instead. For example, one could say, ``Writers abuse adverbs.''
  - Jokes or puns: They have no place in a formal document.
  - Bad, good, nice, terrible, stupid: A scientific dissertation does not make moral judgments.
    - Use ``incorrect/correct'' to refer to factual correctness or errors. Use precise words or phrases to assess quality (e.g., “method A requires less computation than method B”).
    - In general, one should avoid all qualitative judgments.
  - Perfect: Nothing is.

Guidelines

- Terms And Phrases To Avoid:
  - “An ideal solution”: You're judging again.
  - Today, Modern times: Today is tomorrow's yesterday.
  - Soon: How soon? Later tonight? Next decade?
  - We were surprised to learn...: Even if you were, so what?
  - Would seem to show: all that matters are the facts.
  - In terms of‘: usually vague
  - Different: Does not mean “various”: different than what?

Guidelines

- Terms And Phrases To Avoid:
  - You will read about...: The second person has no place in a formal dissertation.
  - I will describe...: The first person has no place in a formal dissertation. If self-reference is essential, phrase it as “Section 10 describes...”
  - We as in we see that: A trap to avoid. Reason: almost any sentence can be written to begin with “we” because “we” can refer to: the reader and author, the author and advisor, the author and research team, experimental computer scientists, the entire computer science community, the science community, or some other unspecified group.

Guidelines

- Terms And Phrases To Avoid:
  - Hopefully, the program...: Computer programs don't hope, not unless they implement AI systems. By the way, if you are writing an AI thesis, talk to someone else: AI people have their own system of rules.
  - ...a famous researcher...: It doesn't matter who said it or who did it. In fact, such statements prejudice the reader.

Guidelines

- Terms And Phrases To Avoid:
    - Be Careful When Using “few, most, all, any, every”: A dissertation is precise.

    If a sentence says “Most computer systems contain X”, you must be able to defend it. Are you sure you really know the facts? How many computers were built and sold yesterday?
    - “must”, “always”: Absolutely?
    - “should”: Who says so?
    - “proof”, “prove”: Would a mathematician agree that it's a proof?
    - “show”: Used in the sense of “prove”. To “show” something, you need to provide a formal proof.
    - “can/may”: Your mother probably told you the difference.

Guidelines

- Voice
  - Use active constructions. For example, say “The operating system starts the device” instead of “the device is started by the operating system.”
- Tense
  - Write in the present tense. For example, say “The system writes a page to the disk and then uses the frame...” instead of “The system will use the frame after it wrote the page to disk...”
- Define Negation Early
  - Example: say “no data block waits on the output queue” instead of “a data block awaiting output is not on the queue.''

Guidelines

- Focus On Results And Not The People/Circumstances In Which They Were Obtained
  - After working eight hours in the lab that night, we realized...'' has no place in the dissertation. It doesn't matter when you realized it or how long you worked to obtain the answer.
  - Another example: “Jim and I arrived at the numbers shown in Table 3 by measuring...''
  - Put an acknowledgement to Jim in the dissertation, but do not include names (even your own) in the main body. You may be tempted to document a long series of experiments that produced nothing or a coincidence that resulted in success. Avoid it completely.
  - In particular, do not document seemingly mystical influences (e.g., “if that cat had not crawled through the hole in the floor, we might not have discovered the power supply error indicator on the network bridge”).
  - Never attribute such events to mystical causes or imply that strange forces may have affected your results.
  - Summary: stick to the plain facts. Describe the results without dwelling on your reactions or events that helped you achieve them

Guidelines

- Avoid Self-Assessment (both praise and criticism)
  - Both of the following examples are incorrect: “The method outlined in Section 2 represents a major breakthrough in the design of distributed systems because...”; “Although the technique in the next section is not earthshaking,...”
- References To Extant Work
  One always cites papers, not authors. Thus, one uses a singular verb to refer to a paper even though it has multiple authors.
  - For example “Johnson and Smith [J&S90] reports that...”Avoid the phrase “the authors claim that X”. The use of “claim” casts doubt on “X” because it references the authors' thoughts instead of the facts. If you agree “X” is correct, simply state “X” followed by a reference. If one absolutely must reference a paper instead of a result, say ”the paper states that...” or “Johnson and Smith [J&S 90] presents evidence that...”.

Guidelines

- Knowledge Vs. Data
  - The facts that result from an experiment are called “data”. The term “knowledge” implies that the facts have been analyzed, condensed, or combined with facts from other experiments to produce useful information.

- Cause and Effect:
  - A dissertation must carefully separate cause-effect relationships from simple statistical correlations.
  - For example, even if all computer programs written in Professor X's lab require more memory than the computer programs written in Professor Y's lab, it may not have anything to do with the professors or the lab or the programmers (e.g., maybe the people working in professor X's lab are working on applications that require more memory than the applications in professor Y's lab).

Guidelines

- Drawing Only Warranted Conclusions:

  One must be careful to only draw conclusions that the evidence supports. For example, if programs run much slower on computer A than on computer B, one cannot conclude that the processor in A is slower than the processor in B unless one has ruled out all differences in the computers' operating systems, input or output devices, memory size, memory cache, or internal bus bandwidth.

  In fact, one must still refrain from judgment unless one has the results from a controlled experiment (e.g., running a set of several programs many times, each when the computer is otherwise idle). Even if the cause of some phenomenon seems obvious, one cannot draw a conclusion without solid, supporting evidence.

Guidelines

- Drawing Only Warranted Conclusions:

  Commerce and Science:

  - In a scientific dissertation, one never draws conclusions about the economic viability or commercial success of an idea/method, nor does one speculate about the history of development or origins of an idea.
  - A scientist must remain objective about the merits of an idea independent of its commercial popularity. In particular, a scientist never assumes that commercial success is a valid measure of merit (many popular products are neither well-designed nor well-engineered). Thus, statements such as “over four hundred vendors make products using technique Y” are irrelevant in a dissertation.

Adapted from: “How To Write A Dissertation”
http://www.cs.purdue.edu/homes/comer/essay.dissertation.html

What about the …

- Title
    - It must catch the readers attention
    - It presents synthetically the idea or major contribution of project
    - Should not be too lengthy
- Abstract
    - Presents a clear explanation of the problem and how it was solved
    - It doesn’t have any reference
- Dissertation Corpus
    - It presents the work carried in the research
    - Usual structure:
        - Introduction
        - State of the Art
        - Development
        - Conclusions and Future Work

What about the …

- Introduction
  - It must present the theme and problem of the research, as well as the goals, approaches and synthesis of the results achieved.
- Introduction (example structure)
  - Motivation:
    - Presents theme and problem, motivating the learner for further reading
  - Context:
    - Describes the context of the research
  - Goals:
    - Presents the general and specific goals
  - Methodology/Development
    - Presents the steps of the development as well as the applied methods
  - Scope
    - Describes the domain and scope of the research and presents the main conclusions
  - Organisation of the document
    - Describes briefly each of the chapters of the report

What about the …

- Introduction
  - Theme should be presented generally, taking the reader to study focus (problem).
  - Should avoid long introductions. It should introduce the reader in the domain and motivate him to read the rest in order to fully understand the work conducted.

What about the …

- State of the Art
  - It is important to be objective in the bibliographic review
  - It should not include all that has been read, only the relevant work for the understanding/comprehension of the work conducted.
  - By including non related information, false expectation are risen in the reader
  - Presents the main concepts of the research domain. Use and reference different sources of information
  - It should present in brief and direct way how the ideas have evolved through time, showing also how the research conducted fits in such research domain.
  - Correlated projects should be presented for further comparison.

What about the …

- Development
    - Chapters in the development should present the work carried out and contributions
    - Needed concepts should have been presented in State of the Art Chapter
    - There is no need to refer again the work from other authors
    - If any comparison is done, it should have been also mentioned in the State of the Art Chapter
    - The comparison should be objective
    - The chapters of the development should present the construction of the theory/model/proposal
    - Concepts created by the student should be presented here.
    - The evaluation of the work and the evidences that the goals have been achieved are presented here (use data, graphics, tests, case studies, formal proofs, etc.)

What about the …

- Development
  - Should avoid writing development chapters as a pure documentation of a computational system
  - If a system was developed with a specific purpose of discovering new knowledge, the focus should be the knowledge discovery and not the system itself.
  - The purpose of a dissertation is to document an original/different contribution to knowledge and not a tool
  - The tool can be used to demonstrate that an original contribution to knowledge was achieved, p.ex., by its usage or by the ideas implemented in it

What about the …

- Conclusions and Future Work
  - It must have:
    - the conclusions themselves;
    - the contributions of the work;
    - Publications;
    - Next steps.
  - In the conclusions, the student should make reference to the tackled problem and its solutions
  - How the solution was achieved
  - Comment about the results obtained in the evaluation
  - Should have a discussion not only about the advantages/strong points but also the weak ones and the limitations.
  - The lessons learnt are also relevant in this chapter

What about the …

- Conclusions and Future Work
  - The contributions should be listed in scale of importance (ending with the major ones!)
  - Publications are not contributions are results and communication of the research
  - A good research project doesn’t end in one contribution, on the opposite it opens new possibilities for further researches.
  - In future work, these opportunities should be pinpointed.

What about the …

- References
  - List of the references mentioned in the body of the dissertation
    - Do not list what was not referenced;
    - Do not cite if you are not listing

What about the …

- The dissertation doesn’t have to be written in the order that is organized;
- The introduction may be the first to be written
- The development can only be written after the conclusion of the research
- Do not finish your State of the Art before concluding the development since new things can still be found
- Nonetheless, Drafts can be always written!
- Conclusion can only be written after the chapters associated with the development are finished
- References are continuously composed as you write your document. Do not leave that part to the end!!

What about the …

- Adopt a unique style of writing
- Academic works are written in the 3rd person of the singular
- It is accepted the 1st person of the plural. Do not mix styles
- Use the present in the whole document
- In the conclusions, the past is accepted
- Background your statements with references
- Do not use jokes or irony

What about the …

- 7 sins of an Academic Writing
  - Long Sentences
  - Orthographic errors
  - Literal translations
  - Illegible Figures/Tables
  - Grammatical Errors
  - Copying without “” or citation
  - Bla bla bla

What about the …

- Other Sins
  - No connection between ideas
  - Too many sections/ Many sub-headings
  - Unconnected Sentences
  - Badly built sentences or confused ideas
  - Ideas from other authors without citation
  - Abuse of English terms

What about the …

- Clues to good quality texts
  - Careful Reading (author, supervisor, and others)
  - Use automatic revisers
  - Divide paragraphs carefully. Each paragraph should present a central idea
  - One section should have more than one paragraph
  - Abbreviations should be introduced, the first time they appear in the text, they should be written comprehensively (even the most common ones like UML)

Adapted from: http://www.inf.ufes.br/~pdcosta/ensino/2010-2-metodologia-de-pesquisa/material/Redacao_Trabalho_Conclusao